MDI3003
ADVANCED PREDICTIVE ANALYTICS

EXPERIMENT 08

Agricultural Predictive Analytics
Rice Yield Regression

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Github link: https://github.com/Shobana0212/Lab8_Agricultural-Predictive-Analytics

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1. Abstract

This report presents a reproducible agricultural predictive-analytics experiment for rice yield regression using district-level historical observations. The primary objective is to develop and evaluate regression models that estimate rice yield from information available before harvest. The modelling target is rice yield expressed in tonnes per hectare (t/ha). Predictive analytics in agriculture can support the analysis of historical production patterns and help identify how yield estimates vary across locations and years. However, the reliability of such predictions depends on the quality of the dataset, the availability of relevant predictors, the selection of appropriate evaluation procedures, and the extent to which the results generalize to future observations. Therefore, this experiment emphasizes a systematic modelling workflow, chronological data separation, consistent model comparison, and transparent reporting of results and limitations.

The experiment uses historical observations organized by state, district, and year. The evaluation procedure follows a chronological boundary to reflect the practical task of predicting yields for later harvest years using earlier observations. All available years before the validation period are assigned to training, the two years immediately preceding the final test period are reserved for validation, and the latest two years are kept exclusively for final testing. Specifically, the validation period covers 2021–2022, while the held-out test period covers 2023–2024. This approach reduces the risk of evaluating a model using future-year information during training. The training data are used for preprocessing, model fitting, and exploratory analysis, while the validation data determine the final candidate model. The test data are used only for the final evaluation after model selection.

Four predeclared regression candidates are evaluated: a training-median baseline, Ridge regression, a fixed-depth Decision Tree, and a Random Forest. The median baseline provides a simple reference point against which the predictive performance of the more complex models can be assessed. Ridge regression provides a regularized linear approach for modelling relationships between the available predictors and rice yield. The Decision Tree captures nonlinear relationships through a sequence of feature-based splits, while the Random Forest combines multiple decision trees to produce predictions. The models are compared using the same training observations and a common validation set. Mean Absolute Error (MAE) is the primary selection metric because it expresses the average absolute prediction error in the original target unit and is less sensitive to individual extreme errors than squared-error-based measures. Root Mean Squared Error (RMSE) and the coefficient of determination (R^2) are used as secondary evaluation metrics.

The preprocessing workflow applies categorical encoding to state and district and standardizes the year variable using transformations fitted on the training data. The categorical encoders use OneHotEncoder(handle_unknown='ignore'), allowing the pipeline to handle categories that were not observed during fitting without failing during prediction. The year variable is processed using median imputation followed by standardization within the training-fitted pipeline. Area and production are excluded from the predictor set because production and area can be used to reconstruct yield, potentially introducing target leakage into the model. Yield recorded in kilograms per hectare is converted to tonnes per hectare by dividing by 1,000. These decisions establish a consistent

prediction target and reduce the risk of using fields that would not be appropriate for the intended pre-harvest prediction task.

The numerical results in this report are reproduced exactly from the submitted result table; no values have been recalculated or replaced. Ridge regression is selected because it achieves the lowest validation MAE of **0.444447 t/ha** among the four evaluated candidates. On the held-out 2023–2024 test period, the selected Ridge trend model records an MAE of **0.388552 t/ha**, an RMSE of **0.635615 t/ha**, and an R^2 of **0.523184**. The median baseline records a test MAE of **0.833910 t/ha**, an RMSE of **1.064220 t/ha**, and an R^2 of **−0.336676**. These results indicate that the selected Ridge model performs better than the median baseline on the reported test metrics. Nevertheless, the results should be interpreted as evidence of predictive performance on the specified test period rather than proof that the model will perform equally well across all future years, regions, or agricultural conditions.

Further evaluation examines the distribution of prediction errors, the five largest nonzero test errors, year-specific test metrics, and rolling-origin performance within the development period. The actual-versus-predicted plot provides a visual comparison between observed and estimated yields, while the residual plot helps identify systematic prediction errors and possible variation in error magnitude. The largest errors include both over-predictions and under-predictions, demonstrating that the selected model does not estimate every district-year observation with equal accuracy. Rolling-origin evaluation provides an additional descriptive assessment of temporal stability by fitting fresh models on earlier years and evaluating them at subsequent development-year origins. Variation between origins indicates that performance depends to some extent on the particular year being evaluated. These checks complement the main validation and test results but do not establish statistical confidence intervals or guarantee future performance.

Several limitations affect the interpretation of this experiment. First, the supplied dataset does not contain a season field, although the laboratory specification calls for season as a categorical predictor. Consequently, the implemented model uses the available state–district–year formulation, and full compliance with the intended season-based schema remains pending. Second, source and licence verification has not been completed, so the provenance and reuse permissions of the exact local dataset require confirmation. Third, the core model does not include observed weather, soil characteristics, irrigation practices, crop variety, or other agricultural variables that may influence rice yield. Their absence may contribute to prediction errors, but the available results cannot establish any of these factors as the cause of a particular error. In addition, the observations are aggregated at district level, and the evaluation focuses on chronological generalization rather than independent spatial generalization to unseen regions.

Overall, the experiment demonstrates a structured approach to agricultural regression modelling, including data-quality checks, chronological evaluation, training-only preprocessing, baseline comparison, model selection using validation MAE, final test evaluation, error analysis, and rolling-origin robustness checks. Ridge regression is the selected candidate under the predefined validation rule, and its reported test performance improves on the median baseline. The findings support the use of the fitted model for analysis within the limits of the available dataset and evaluation design.

Before operational deployment, the dataset provenance and licence should be verified, the missing season field should be addressed with a corrected dataset or instructor-approved deviation, and additional agricultural predictors and broader temporal and spatial validation should be considered. These steps would help establish whether the observed performance can be maintained under different agricultural conditions and future harvest periods.

2. Introduction

Agricultural yield prediction is an important application of predictive analytics in which the response variable is continuous and the objective is to estimate the expected crop yield for a specified geographic location and harvest period. Accurate yield estimation can support agricultural planning, resource allocation, production assessment, and the identification of variations in crop performance across regions and years. However, agricultural yield is influenced by several interacting factors, including climatic conditions, soil properties, irrigation practices, crop varieties, and cultivation methods. Developing a reliable prediction model therefore requires appropriate data preparation, careful feature selection, and an evaluation strategy that reflects the intended prediction scenario.

This experiment focuses on rice yield regression using district-level historical observations. The experiment emphasizes the development of a reproducible machine-learning workflow, comparison of regression algorithms, evaluation of prediction errors, and assessment of model performance on later harvest years. Rather than relying only on random train-test splitting, the experiment adopts chronological evaluation to examine whether models trained on historical observations can predict yields in subsequent years. This design is particularly relevant to agricultural prediction because information from future harvest periods should not be available during model training or model selection.

The analysis follows a chronological data-partitioning strategy. Earlier observations are assigned to the training set, the two years immediately preceding the test period are used for validation, and the latest two observed years are reserved for final testing. The validation set is used to compare candidate models and select the model with the lowest Mean Absolute Error (MAE), while the test set provides a final assessment of the selected model and a predeclared baseline. This separation helps prevent test-set results from influencing feature selection or model settings and provides a clearer estimate of performance on later observations. The experiment also includes year-wise error analysis and rolling-origin evaluation to examine whether performance varies across different temporal batches.

Four regression approaches are considered: a training-median baseline, Ridge regression, a fixed-depth Decision Tree, and a Random Forest. The median baseline provides a simple reference for evaluating whether machine-learning models offer practical improvement. Ridge regression represents a regularized linear approach, while the Decision Tree and Random Forest allow more flexible modelling of nonlinear relationships. The candidates are evaluated on common data partitions using MAE as the primary selection metric, with Root Mean Squared Error (RMSE) and the

coefficient of determination (R^2) reported as secondary metrics. Actual-versus-predicted and residual plots are also used to examine the relationship between observed yields and model estimates and to identify substantial prediction errors.

A central consideration in this experiment is the prevention of target leakage. The dataset contains production and harvested-area fields, but these variables are excluded from the model predictors. Since yield can be reconstructed from production divided by harvested area, including these fields could provide information that would not be appropriate for a prospective pre-harvest prediction task. They are therefore retained as audit fields rather than used as model inputs. The implemented predictor set consists of state, district, and year. Categorical variables are encoded within the modelling pipeline, and year is standardized using transformations fitted on the training data. Keeping preprocessing within the pipeline helps ensure that validation and test observations do not influence the learned preprocessing parameters.

The analysis is intentionally conservative in its interpretation. Although state and district identify geographic context and year represents temporal information, these variables do not directly describe the agronomic conditions that determine crop yield. The resulting model should therefore be understood as a limited prospective-design backtest based on the available historical dataset, rather than a complete agricultural forecasting system. Its performance may be affected by unobserved weather patterns, soil characteristics, irrigation availability, crop varieties, changes in agricultural practices, and inconsistencies in district-level reporting. These factors are plausible explanations for prediction errors, but the experiment does not establish them as causal explanations.

An additional dataset-specific limitation is the absence of a season field. The laboratory specification calls for season to be included among the categorical predictors, but the supplied dataset does not provide this variable. The current implementation consequently uses state, district, and year, and the missing season field must be treated as a documented deviation rather than replaced with invented values. In addition, the exact dataset source and licence remain subject to verification. These limitations affect the completeness and reproducibility of the intended experiment and should be resolved, or explicitly approved by the instructor, before the model is considered fully compliant with the laboratory specification.

The main objectives of this experiment are to establish a clear prediction contract, validate the dataset structure, inspect data quality, create chronological data partitions, implement training-only preprocessing, compare regression candidates using consistent evaluation criteria, select a model using validation MAE, and assess its performance on the untouched test period. Further objectives include examining large prediction errors, comparing metrics across the two test years, evaluating temporal robustness through rolling origins, and verifying that the saved model reproduces its predictions after reloading. Together, these steps provide a structured basis for evaluating the suitability and limitations of the selected rice yield regression model.

3. Objectives

- Define the prediction target, observation unit, prediction horizon and feature-availability boundary.
- Audit the uploaded rice dataset for duplicates, missingness, invalid targets and repeated state–district–year keys.
- Construct a chronological train/validation/test split without using held-out target distributions for feature decisions.
- Fit categorical encoding and numerical imputation/scaling only on training data.
- Compare a median baseline, Ridge regression, Decision Tree and Random Forest using MAE, RMSE and R^2.
- Select the model using validation MAE and evaluate it once on the untouched test years.
- Inspect the largest errors, year-wise robustness and reload reproducibility.
- Document limitations, data-governance gaps and responsible interpretation.

4. Problem Statement and Prediction Contract

4.1 Problem Statement

Can historical district-level state, district, and harvest-year information be used to predict rice yield for later harvest years? This experiment addresses the problem using regression models trained on historical rice-yield observations. The objective is to estimate rice yield for future harvest years using only the permitted geographic and temporal predictors. The model is evaluated using chronological data partitions to assess its ability to generalize to later years without using future observations during training or model selection.

4.2 Prediction Contract

The prediction contract defines the observation unit, target variable, measurement unit, crop scope, prediction horizon, permitted predictors, and excluded fields. Establishing these elements before modelling ensures that the task remains consistent throughout data preparation, model training, and evaluation.

Contract item	Definition
Observation unit	One rice-producing district–harvest-year observation in the uploaded dataset.
Target	Rice yield, represented as yield_t_ha.
Target unit	Tonnes per hectare (t/ha).
Crop scope	Rice records in the supplied dataset.
Prediction horizon	Before the target harvest, using only the permitted state, district and year inputs.
Predictors used	State, district and year.
Excluded post-harvest fields	Area (Ha) and Production (Tonnes).
Missing required field	Season is absent from the supplied dataset; no season values were invented.

4.3 Prevention of Target Leakage

Production and harvested area are excluded from the predictor set because rice yield can be derived from production divided by harvested area, with the appropriate unit conversion. Including production as an input would provide information closely related to the target that is not available under the intended pre-harvest prediction setting. This could produce misleading evaluation results by allowing the model to exploit post-harvest information instead of learning from the permitted predictors.

Therefore, Area (Ha) and Production (Tonnes) are retained only for data auditing and provenance checks. The model uses state, district, and year as predictors, with rice yield expressed in t/ha as the target variable.

4.4 Dataset-Specific Deviation

The laboratory specification requires Season as a categorical predictor, but the supplied dataset does not contain a season field. Consequently, season cannot be included in the current implementation without obtaining additional source data. No season values have been inferred or fabricated. This deviation must be documented, and instructor approval or a corrected dataset is required to establish full compliance with the intended schema.

The prediction contract is therefore limited to the available state–district–year formulation. The model's results should be interpreted within this scope and should not be taken as evidence that seasonal effects have been captured.

5. Dataset Description and Data Governance

5.1 Dataset Overview

The experiment uses the dataset Rice_Data_2026-10-12.csv, which contains historical district-level observations for rice yield prediction. The dataset provides geographic and temporal information used to estimate rice yield in tonnes per hectare (t/ha). The raw dataset contains 15,735 rows, of which 15,717 rows were retained for modelling after the documented data-cleaning procedures. The observations cover the period from 1988 to 2024, with 30 unique observed years.

The training data contain 36 distinct states and 734 distinct district names. These geographic categories provide the spatial identifiers used by the model, while the year variable represents the temporal dimension. The dataset is used to construct chronological training, validation, and test partitions so that the model is evaluated on later harvest years.

5.2 Dataset Summary

Attribute	Recorded result
Dataset file	Rice_Data_2026-10-12.csv
Raw rows	15,735
Retained modelling rows	15,717
Observed years	1988–2024; 30 unique observed years
Training states	36

Training districts	734
Raw SHA-256	74a017708b0c084600b334bec644e11c73817438a7e0786c86498eb632d5a2b9
Source / licence status	Publisher/licence not verified in the notebook result record
Important schema deviation	Season column is absent; instructor approval remains pending.

5.3 Data Integrity and Auditability

Data integrity is important because errors in the input dataset can affect model training, evaluation, and reproducibility. The raw-file SHA-256 hash is retained as an audit identifier for the supplied dataset. A cryptographic hash can be used to check whether a file matches the version associated with the recorded identifier. It does not, by itself, establish the dataset's publisher, authenticity, or licence.

The data-cleaning process removed 13 exact duplicate rows and excluded five rows associated with two conflicting state–district–year key groups. After conflict removal, no duplicate state–district–year keys remained, and no invalid target, geographic, or year records were reported. Seventeen rows with missing production values were retained for audit, and 24 zero-yield rows were retained rather than automatically discarded. These decisions preserve the documented treatment of the dataset and avoid silently changing observations solely because their values may appear unusual.

5.4 Data Governance and Licensing

The notebook result record does not contain completed verification of the dataset publisher, exact source URL, or applicable licence. Therefore, this report does not claim that the dataset's external source or reuse permission has been verified. Before the dataset is redistributed or used beyond the intended laboratory exercise, the original download source and its access conditions should be confirmed and documented.

The raw dataset should be preserved unchanged as the reference input. Cleaning and modelling transformations should be performed separately so that the original observations remain available for auditing. The recorded hash should be checked against the exact raw file used in the experiment, and the dataset version, source, access date, and licence evidence should be added once verified. This practice supports reproducibility and helps distinguish observed data from processing decisions.

5.5 Schema Limitation

A significant limitation is that the supplied dataset does not contain a Season column, although the laboratory specification requires season to be used as a categorical predictor. The current model therefore uses state, district, and year only. Season values have not been fabricated or inferred from unsupported assumptions. The missing field must be addressed by obtaining a corrected dataset or securing instructor approval for the documented deviation.

In summary, the dataset provides a substantial collection of historical district-level rice-yield observations suitable for the implemented state–district–year regression experiment. However, complete compliance with the intended schema and data-governance requirements remains conditional on resolving the missing season field and verifying the exact dataset source and licence.

6. Data Integrity and Canonical Preparation

6.1 Data Cleaning and Integrity Checks

Data integrity is essential for reliable regression modelling because duplicate records, conflicting observations, invalid values, and inconsistent units can affect model training and evaluation. Before fitting the models, the supplied rice dataset was subjected to data-quality checks to identify duplicate rows, conflicting geographic-temporal keys, invalid records, missing production values, and zero-yield observations. The cleaning decisions were documented to ensure that the resulting dataset could be audited and reproduced.

6.2 Data Integrity Audit

Audit item	Result / decision
Exact duplicate rows	13 removed.
Conflicting state–district–year key groups	5 rows from 2 conflicting key groups excluded.
Invalid target/geography/year rows after conflict removal	0.
Missing production values	17 retained for audit.
Zero-yield rows	24 retained.
Duplicate state–district–year keys after cleaning	0.
Target conversion	Yield (Kg/ha) divided by 1,000 to obtain t/ha.
Leakage control	Area and Production excluded from predictors.

6.3 Duplicate and Conflict Handling

Exact duplicate rows were removed to prevent identical observations from being counted more than once. The audit also identified conflicting observations sharing the same state–district–year key. Five rows belonging to two conflicting key groups were excluded because these records did not provide a unique observation for the corresponding geographic and temporal combination. After these removals, the cleaned data contained no duplicate state–district–year keys.

The integrity checks also reported zero invalid target, geographic, or year records after conflict removal. Seventeen records with missing production values were retained for audit rather than being silently discarded. Similarly, 24 zero-yield observations were retained. These decisions preserve the documented treatment of potentially important records and avoid introducing undocumented filtering based solely on unusual values.

6.4 Canonical Target Preparation

The target variable is standardized to a consistent measurement unit before modelling. The original yield values are recorded in kilograms per hectare (Kg/ha), whereas the regression task requires tonnes per hectare (t/ha). Therefore, the target is converted by dividing yield in kilograms per hectare by 1,000.

This conversion ensures that all reported dimensional error metrics, including Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), are expressed in t/ha. Maintaining consistent units is important when comparing model performance and interpreting the magnitude of prediction errors.

6.5 Leakage Prevention and Feature Preparation

Area (Ha) and Production (Tonnes) are excluded from the predictor set because production and harvested area can be used to reconstruct yield, subject to the appropriate unit conversion. Including these fields as inputs would introduce post-harvest information into a prediction task framed around pre-harvest estimation. Such leakage could make the model appear more accurate than it would be when only permitted predictors are available.

Consequently, the predictive feature set consists of state, district, and year. Area and production remain outside the model inputs and are used only for data auditing where appropriate. This separation maintains a clear distinction between the information available for prediction and the fields associated with the observed outcome.

6.6 Canonical Preparation Limitation

The cleaned canonical data preserve the target in a dimensional form suitable for regression and maintain the recorded data-integrity decisions. However, the supplied dataset does not contain the required Season field. No season values were fabricated or inferred merely to make the schema appear complete. The current implementation therefore remains limited to the available state–district–year structure, and the missing field must be addressed through a corrected dataset or documented instructor approval.

Overall, the data-integrity and canonical-preparation stage establishes a consistent target unit, removes duplicate and conflicting records, documents retained observations, and prevents the use of area and production as predictive inputs. These steps provide the foundation for the chronological evaluation and regression-model comparison described in the following sections.

7. Exploratory Data Analysis

7.1 Overview

Exploratory Data Analysis (EDA) was performed using the training dataset to understand the distribution of rice yield, the coverage of harvest years, and the frequency of district observations. The analysis was restricted to training data so that the held-out validation and test target distributions would not influence feature selection or modelling decisions. This approach maintains the chronological evaluation boundary and reduces the risk of information leakage during model development.

The training target distribution was examined using descriptive statistics, including the mean, standard deviation, minimum, quartiles, median, and maximum. These statistics provide an overview of the typical yield values and the degree of variation within the training observations.

7.2 Training Target Distribution

Training EDA statistic	Value
Mean yield (t/ha)	2.108491
Standard deviation (t/ha)	2.165889
Minimum (t/ha)	0.000000
Q1 (t/ha)	1.350000
Median (t/ha)	2.030000
Q3 (t/ha)	2.700000
Maximum (t/ha)	223.130000
Validation-only states	0
Validation-only districts	33

The mean training yield is 2.108491 t/ha, while the median is 2.030000 t/ha. The first and third quartiles are 1.350000 t/ha and 2.700000 t/ha, respectively. The maximum yield of 223.130000 t/ha is substantially higher than the central quartiles. This indicates a highly uneven target distribution with an extreme upper-tail observation or observations that warrant data-quality awareness during interpretation.

Such a large maximum relative to the quartiles can affect squared-error metrics such as RMSE more strongly than absolute-error metrics such as MAE. Therefore, both metrics are retained for model evaluation so that average absolute errors and the influence of large errors can be considered separately. The recorded statistics are descriptive and do not, by themselves, establish the cause of the extreme value.

7.3 Geographic Coverage and Potential Bias

The training data contain 36 states and 734 district names. The category audit found zero validation-only states and 33 validation-only district names. Thus, although the validation data do not introduce new state categories, some district names occur in validation without appearing in the training category set.

Two observations emerge from the training-only EDA:

1. **Uneven yield distribution:** The maximum value of 223.130000 t/ha is far above the first and third quartiles, indicating a long upper tail relative to the central distribution. Large errors associated with extreme observations may influence RMSE and should be examined individually.
2. **Changing district coverage:** There are 33 validation-only district names. The model must therefore handle categorical values that were not present during training. The selected encoder configuration supports this by ignoring unseen categories during transformation.

A potential source of bias is that district-year observations may reflect changes in administrative boundaries, reporting practices, or unobserved agricultural conditions. Since these factors are not directly represented in the predictor set, their influence cannot be separated from model error using the available data alone.

8. Chronological Evaluation Design

8.1 Data Partitioning Strategy

The experiment uses chronological partitioning to evaluate the model's ability to predict rice yield for later harvest years. Unlike a random split, this design preserves the temporal ordering of observations and avoids training on years later than those used for validation or testing.

The latest two observed years are reserved for testing, while the preceding two years are used for validation. All earlier available years are assigned to training. The validation set supports model comparison and selection, whereas the test set is reserved for the final evaluation of the selected model and the predeclared median baseline.

8.2 Split Manifest

Partition	Years	Rows	Unique districts
Train	1988; 1996–2020 (all available years)	13,457	734
Validation	2021, 2022	1,324	678
Test	2023, 2024	936	701

The training partition contains 13,457 observations, the validation partition contains 1,324 observations, and the test partition contains 936 observations. These partitions follow the documented year-based split, with no overlap in their assigned years.

The training period includes the available observations from 1988 and 1996–2020. The missing intervening years are not treated as observed years. Validation uses 2021 and 2022, and the final test uses 2023 and 2024.

8.3 Evaluation Boundary and Test Protection

All model settings and feature decisions are determined using the development data and the predefined validation procedure. The selected model is not refitted on combined training and validation data before testing. This preserves the specified evaluation design and ensures that the final test scores correspond to the model selected using training-only fitting and validation MAE.

The test period is opened only for the final evaluation. Test performance is not used to modify model settings, change the feature set, or select a different model. The year-wise test metrics are used descriptively to assess variation between the two test-year batches rather than to establish a confidence interval.

9. Preprocessing Pipeline

9.1 Overview

Preprocessing is implemented within the modelling pipeline so that transformations are learned only from training observations. This prevents validation and test data from influencing the learned

category sets or numerical scaling parameters. The same fitted transformations are then applied to later observations during validation and testing.

The predictor set consists of state, district, and year. Area and production are excluded because they are post-harvest or audit fields that could introduce target leakage. The required Season field is not available in the supplied dataset and is therefore not encoded.

9.2 Feature Transformations

Feature	Transformation	Fit subset	Justification
State	OneHotEncoder(handle_unknown='ignore')	Training only	Categorical geographic input available before harvest.
District	OneHotEncoder(handle_unknown='ignore')	Training only	District history is available before harvest; unseen validation categories are ignored.
Year	Median imputation → StandardScaler	Training only	Numeric harvest-year index; imputation is defensive.
Season	Not available / not encoded	N/A	The supplied dataset has no season column.
Area (Ha)	Excluded	N/A	Post-harvest/audit field; excluded to avoid leakage.
Production (Tonnes)	Excluded	N/A	Post-harvest field that can reconstruct yield with area.

9.3 Categorical Encoding

State and district are encoded using OneHotEncoder(handle_unknown='ignore'). During fitting, the encoder learns categories from the training observations only. This is important because the validation set contains 33 district names that do not appear in the training category set.

The encoder's unknown-category handling allows these validation observations to be transformed without introducing new categories into the fitted encoder. As a result, validation-only district names do not enter the learned training category set. However, this technical handling does not guarantee accurate predictions for unseen districts; it only ensures that the preprocessing pipeline can process them.

9.4 Numerical Preprocessing

Year is processed through median imputation followed by StandardScaler, with both transformations fitted using training rows only. Because the dataset schema requires complete years, the imputer is defensive rather than a substitute for valid temporal information.

Standardizing year places the numeric trend on a comparable scale for Ridge regression and gives its regularization a meaningful feature scale. This monotonic transformation preserves the ordering of years and does not change the ordering used by tree-based split logic.

9.5 Leakage Control and Pipeline Fitting

The complete preprocessing and regression pipeline is fitted only on training observations. Validation and test observations are transformed using the fitted pipeline without recalculating imputation values, category sets, or scaling parameters.

Area (Ha) and Production (Tonnes) remain excluded from the model predictors. This ensures that the model is evaluated using the permitted pre-harvest state, district, and year inputs rather than information that could reconstruct the target after harvest.

In summary, the preprocessing pipeline maintains a consistent transformation process across all model candidates and respects the chronological evaluation boundary. The handling of unseen district categories supports robust pipeline execution, while the absence of Season remains a documented limitation requiring a corrected dataset or instructor-approved deviation.

10. Models and Experimental Settings

Four regression candidates were declared before final testing. The models were selected to provide a comparison between a simple statistical baseline, a regularized linear model, a bounded nonlinear model, and a tree-based ensemble. All candidates use the same training observations and are evaluated using the same validation set.

Model	Predeclared parameters
Median	DummyRegressor(strategy='median')
Ridge trend	Ridge(alpha=1.0, solver='lsqr')
Tree	DecisionTreeRegressor(max_depth=6, min_samples_leaf=10, random_state=42)
Forest	RandomForestRegressor(n_estimators=60, max_depth=12, min_samples_leaf=5, n_jobs=2, random_state=42)

Before fitting, Ridge regression was expected to represent a smooth year trend more naturally than a tree-based model because standardized numeric year can contribute a linear temporal component. This expectation was a hypothesis rather than a predetermined conclusion. The final selection was based on validation MAE, as specified in the evaluation protocol.

11. Validation Results and Model Selection

The four candidates were compared using validation MAE as the primary metric, with RMSE and R^2 as secondary metrics. Runtime was also recorded to compare the computational cost of fitting and validation prediction.

Model / parameters	Validation MAE	RMSE / R^2	Runtime (s)
Median DummyRegressor(strategy='median')	0.850264	1.138439 / -0.430550	0.100521
Ridge trend Ridge(alpha=1.0, solver='lsqr')	0.444447	0.779771 / 0.328854	0.148444
Tree DecisionTreeRegressor(max_depth=6, min_samples_leaf=10, random_state=42)	0.599680	0.854156 / 0.194699	0.134515

Forest RandomForestRegressor(n_estimators=60, max_depth=12, min_samples_leaf=5, n_jobs=2, random_state=42)	0.501631	0.768083 / 0.348823	2.921410

Ridge trend was selected because it achieved the lowest validation MAE of **0.444447 t/ha**. The selection rule was minimum validation MAE, with exact ties retaining the earlier, simpler candidate. No refitting on combined training and validation data was performed after selection.

The Random Forest achieved a lower validation RMSE than Ridge, with values of 0.768083 and 0.779771 respectively. However, the predeclared selection rule prioritizes MAE, so the forest was not selected solely because of its lower RMSE. The forest also had the largest listed runtime at 2.921410 seconds, compared with 0.148444 seconds for Ridge. Under this evaluation rule, Ridge provides the best validation MAE with a substantially lower recorded runtime than the forest.

12. Final Test Evaluation

Model	Test MAE (t/ha)	Test RMSE (t/ha)	Test R^2
Selected Ridge trend	0.388552	0.635615	0.523184
Median baseline	0.833910	1.064220	-0.336676

The selected Ridge trend model achieved a test MAE of 0.388552 t/ha, an RMSE of 0.635615 t/ha, and an R^2 of 0.523184. The median baseline recorded a test MAE of 0.833910 t/ha, an RMSE of 1.064220 t/ha, and an R^2 of -0.336676.

These results indicate that the selected model performs better than the median baseline on all three reported test metrics. The positive R^2 indicates that the selected model explains a meaningful portion of the observed test variance relative to a constant-mean reference. However, this result does not establish causal explanatory power or guarantee equivalent accuracy in future harvest years.

12.1 Actual-versus-Predicted Diagnostic

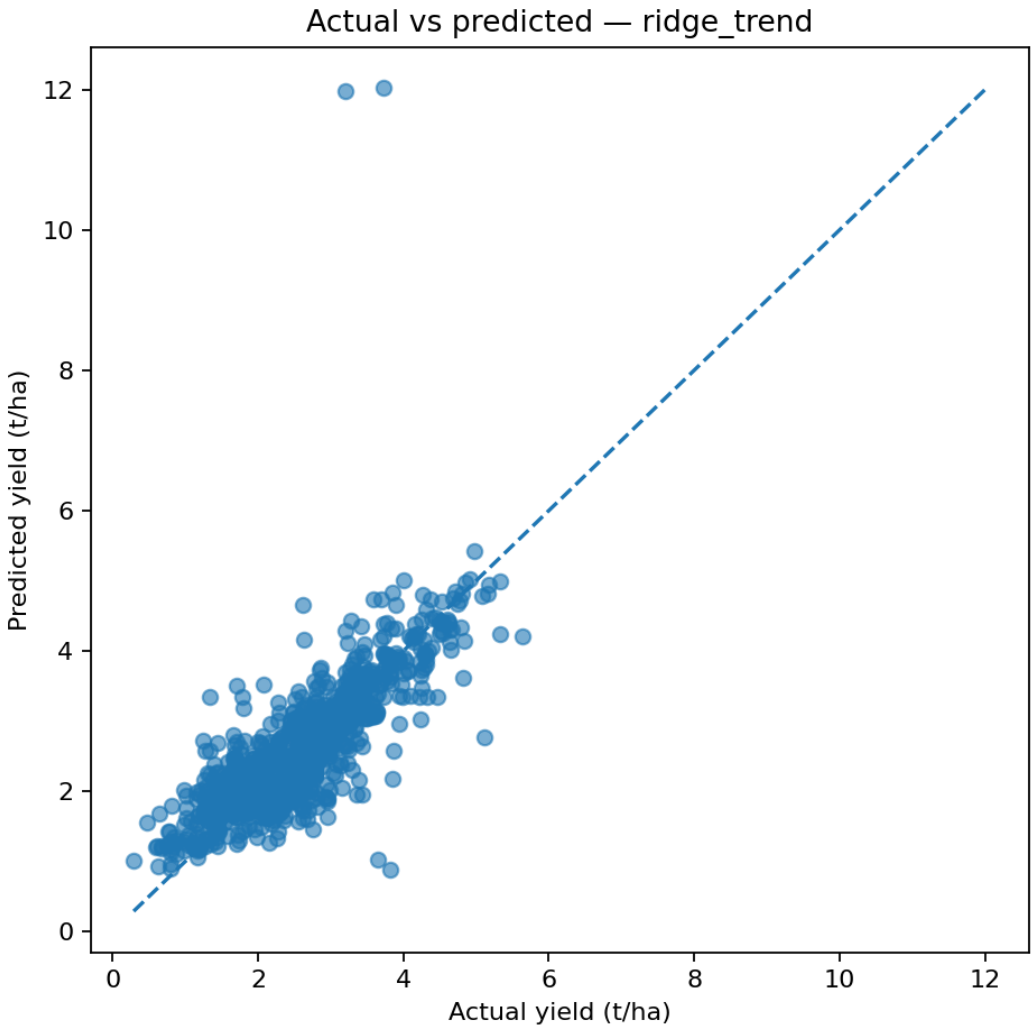


Figure 1. Actual versus predicted rice yield for the selected Ridge trend model on the test set. The dashed line represents equality between actual and predicted yield.

The actual-versus-predicted plot shows a dense central cloud around the equality line, indicating that many predictions are relatively close to their observed values. Several observations are visibly separated from this central group. Two large predicted values near 12 t/ha correspond to the largest negative residual cases identified in the error analysis.

These observations demonstrate why pooled performance metrics should be accompanied by case-level error inspection. A model can achieve a favourable average metric while still producing substantial errors for particular district-year observations.

12.2 Residual Diagnostic

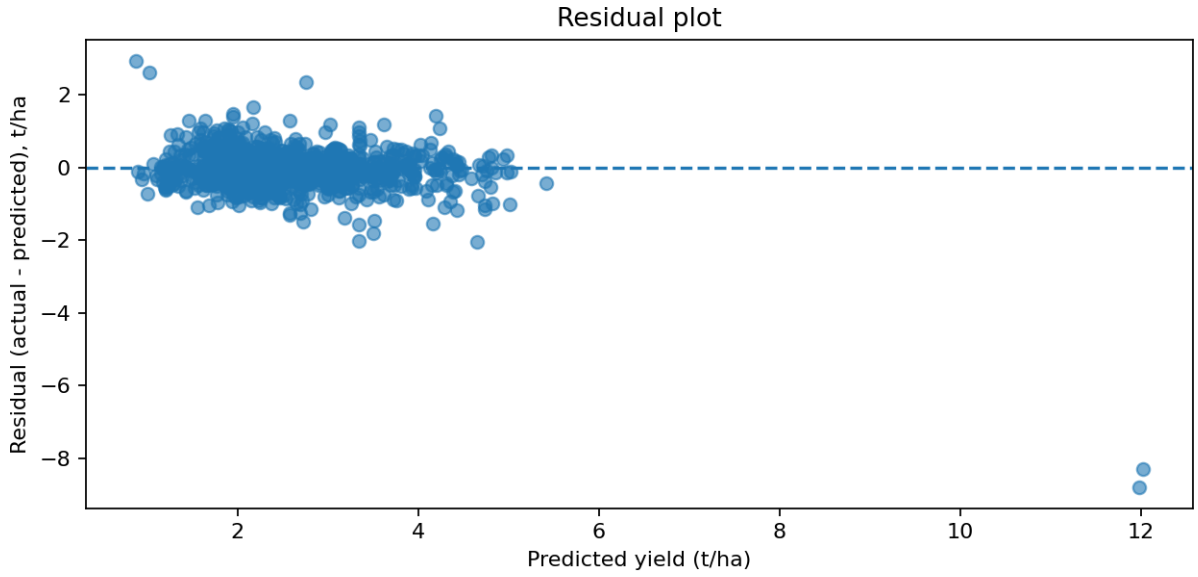


Figure 2. Residuals (actual − predicted) against predicted rice yield for the selected Ridge trend model.

Residuals are calculated as actual yield minus predicted yield. Positive residuals indicate underprediction, while negative residuals indicate overprediction. Most observations lie near the zero-residual line, but a small number of extreme residuals contribute substantially to the visible vertical spread.

The plot supports further inspection of large errors rather than assuming that one explanation applies to every case. Potential explanations remain hypotheses because the dataset does not contain sufficient agronomic and reporting metadata to establish their causes.

13. Year-wise Robustness Analysis

Year	Count	MAE (t/ha)	RMSE (t/ha)	R^2
2023	696	0.398681	0.621165	0.520862
2024	240	0.359178	0.675775	0.528689

The 2023 test batch contains 696 observations, while 2024 contains 240. The 2024 batch has a lower MAE of 0.359178 t/ha than 2023's 0.398681 t/ha, but its RMSE is higher at 0.675775 t/ha compared with 0.621165 t/ha in 2023. This indicates that average absolute error and sensitivity to larger errors do not move identically across the two years. The comparison is descriptive; two test years are insufficient for a temporal confidence interval.

14. Error Analysis

The five largest nonzero test errors are listed below. The suspected causes are hypotheses only; the available dataset does not contain enough agronomic or reporting metadata to confirm any one explanation.

Case	State	District	Year	Actual	Prediction	Error	Abs. error
Case 1	MAHARASHTRA	KOLHAPUR	2023	3.20	11.984499	-8.784499	8.784499
Case 2	MAHARASHTRA	KOLHAPUR	2024	3.73	12.020443	-8.290443	8.290443
Case 3	HARYANA	MAHENDRAGARH	2023	3.82	0.879523	2.940477	2.940477
Case 4	TAMIL NADU	CHENNAI	2023	3.65	1.025773	2.624227	2.624227
Case 5	HIMACHAL PRADESH	SOLAN	2023	5.11	2.762352	2.347648	2.347648

Hypothesis	Interpretation / mitigation
Missing agronomic and environmental variables	Irrigation, rainfall, soil, temperature and water availability are absent; future validated feature enrichment could test this hypothesis.
District boundary or naming changes	Historical geography may not be perfectly comparable; source metadata and boundary history should be checked.
Reporting revisions or anomalies	Unexpected observations may reflect reporting, rounding or revision effects; inspect the original source record.
Variety and management differences	Rice variety, planting practice and input use are not represented; these may create within-district variation.
Sparse district history	Limited observations for some locations can make categorical effects unstable; minimum-history rules could be evaluated in future work.

The two Kolhapur cases are overpredictions because their residuals are negative. The Mahendragarh, Chennai and Solan cases are underpredictions because their residuals are positive. These labels describe the direction of model error only; they do not establish why the observations were difficult to predict.

15. Rolling-origin Robustness

Origin	Training max year	Training rows	Origin rows	MAE	RMSE	R^2	Runtime (s)
2020	2019	12787	670	0.444780	0.939302	0.229268	0.155341
2021	2020	13457	648	0.374308	0.604801	0.511283	0.104018
2022	2021	14105	676	0.501264	0.903348	0.208119	0.094726

The rolling-origin exercise fixes the selected Ridge model family and fits fresh pipelines using only years earlier than each origin. The three recorded MAEs are 0.444780, 0.374308 and 0.501264 t/ha for origins 2020, 2021 and 2022 respectively. The reported mean MAE is 0.4401175001922841 with a range from 0.3743082050835556 to 0.501263998726203. The listed runtime sum is 0.354085 s. These checks diagnose temporal stability; three origins do not establish confidence intervals.

16. Reproducibility and Acceptance Checks

Acceptance item	Recorded result
Raw SHA-256	74a017708b0c084600b334bec644e11c73817438a7e0786c86498eb632d5a2b9
Raw rows	15,735
Canonical rows	15,717
Unique years	30
Split years disjoint	True
Finite, nonnegative target	True
Duplicate keys after cleaning	0
Selected model	ridge_trend
Test lock exists	True
Reload predictions exact	True
Maximum absolute reload difference	0.0
Season deviation approval	PENDING

The selected pipeline was reloaded and produced exactly identical test predictions, with a maximum absolute difference of 0.0. The recorded test lock supports the intended one-time test boundary. The raw-file hash and row counts provide audit identifiers for the dataset version used in the experiment.

The season deviation remains an unresolved acceptance item because the supplied dataset does not contain the required field. In addition, source and licence verification must be completed before the dataset can be treated as fully documented for redistribution or operational use.

17. Limitations and Responsible Interpretation

- Season is absent from the uploaded dataset, so the experiment cannot claim season-specific prediction despite the canonical manual schema.
- Source provenance and licence/access permission were not verified in the notebook record and must be completed before formal redistribution or deployment.
- Area and production are excluded from predictors, so the model cannot use their information to explain yield variation.
- No observed weather, soil, irrigation, water-availability or variety variables are available in the core feature set.
- District boundary changes, repeated locations and reporting anomalies may affect temporal comparability.
- The experiment evaluates temporal generalization but does not assess spatial generalization to unseen districts.
- Two test years provide descriptive year-wise robustness only; they do not support a credible temporal confidence interval.
- Model settings are fixed and no test-set tuning is permitted.
- Feature associations should not be interpreted as causal explanations.

The model should therefore be treated as an academic predictive benchmark rather than an operational agronomic decision system. A prediction can be useful for studying historical patterns

without being sufficient for farm-level recommendations, resource allocation or agronomic intervention.

18. Discussion

18.1 Baseline suitability

The training median of 2.03 t/ha is an appropriate baseline for MAE because the median minimizes the sum of absolute deviations among constant predictors. This provides a robust reference against which learned models can be judged.

18.2 Ridge versus tree and forest

Ridge achieves the lowest validation MAE at 0.444447 t/ha. The Decision Tree and Random Forest have validation MAEs of 0.599680 and 0.501631 t/ha respectively. The forest does improve validation RMSE relative to Ridge, but the predeclared primary metric is MAE, so the lower forest RMSE does not override the selection rule.

18.3 Practical interpretation

The selected model's test MAE of 0.388552 t/ha should be read as an average absolute prediction error over the two held-out years, not as an agronomic guarantee. The presence of several large individual residuals demonstrates why pooled metrics must be accompanied by case-level inspection.

18.4 Responsible use

The experiment does not establish causal relationships between district, year and yield. In particular, the absence of irrigation, weather, soil, variety and management information means that a large residual cannot be assigned to one missing factor without additional evidence. The report therefore uses hypothesis language for error explanations and recommends source validation before any operational use.

19. Conclusion

This experiment presented an end-to-end predictive analytics workflow for agricultural yield prediction using district-level rice production data. The primary objective was to develop and evaluate regression models capable of predicting rice yield in tonnes per hectare using geographical information and historical year data. The study emphasized data quality, leakage prevention, chronological validation, reproducibility, and reliable model evaluation rather than relying only on model complexity.

The dataset initially contained 15,735 records. After removing exact duplicate records and excluding conflicting state–district–year groups, 15,717 records were retained for modeling. The target variable was derived from yield in kilograms per hectare and converted into tonnes per hectare. Production

and area fields were excluded from the predictor set because they are related to post-harvest information and could introduce target leakage. The available predictors were state, district, and year. Since the season field was unavailable, it could not be incorporated into the model, and approval for this deviation remained pending.

A chronological data-splitting strategy was implemented to reflect practical prediction conditions. Earlier observations were used for training, 2021–2022 for validation, and 2023–2024 for final testing. Preprocessing operations, including categorical encoding, year imputation, and scaling, were fitted using only the training data. Four regression approaches were evaluated: a median baseline, Ridge regression, Decision Tree regression, and Random Forest regression. Ridge regression was selected based on the lowest validation mean absolute error (MAE), without using the test results for model selection.

The selected Ridge model achieved a test MAE of **0.388552 t/ha**, an RMSE of **0.635615 t/ha**, and an (R^2) score of **0.523184**. In comparison, the median baseline recorded an MAE of 0.833910 t/ha, an RMSE of 1.064220 t/ha, and an (R^2) score of −0.336676. These results indicate that Ridge regression performed substantially better than the baseline on the held-out test period. The model explained approximately 52.32% of the observed variation in the test data, demonstrating useful predictive capability while leaving considerable variation unexplained.

Further evaluation included separate performance measurements for 2023 and 2024, rolling-origin validation across three historical origins, actual-versus-predicted plots, residual analysis, and examination of the five largest prediction errors. Although the selected model performed consistently across the two test years, certain districts showed substantial prediction errors. These errors may be associated with unusual yield observations, geographical differences, changes in agricultural practices, or the absence of relevant environmental predictors. These explanations remain hypotheses and require further investigation.

The experiment also highlighted important limitations. The dataset did not provide season, weather, soil, irrigation, or crop-variety information. The model relied primarily on location and year, limiting its ability to capture detailed agricultural conditions. Dataset provenance and licensing also require verification. Moreover, the evaluation supports temporal performance on the available observations but does not establish reliable generalization to entirely new geographical regions or demonstrate causal relationships.

In conclusion, the experiment demonstrates that a carefully designed and reproducible regression pipeline can provide meaningful rice yield predictions from historical district-level data. The results support the use of Ridge regression as the selected model for this experiment, subject to the stated limitations and pending compliance checks. Future improvements should include verifying the dataset source and licence, obtaining the missing season information, incorporating weather and soil-related variables, investigating high-error districts, and evaluating the model on additional years and unseen regions. These enhancements could improve predictive accuracy, robustness, and practical usefulness for agricultural planning and decision-making.